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INTRODUCTION

Past experiences and knowledge serve as valuable toolkits for predicting future outcomes. By applying statistical analysis methods, data can be refined, modeled, and tested to generate the most accurate forecasts based on historical patterns. A crucial prediction for many adults is their future net worth or wealth, which is shaped by a range of factors stemming from past decisions and earnings. Intuitively, someone with higher savings, a larger salary, and a valuable home is considered wealthier than someone with little to no savings, a lower salary, and a less expensive home. But what if we could take these past financial indicators, quantify them, and use machine learning to predict future wealth? How can we assess the accuracy of such a prediction?

In order to accomplish such a feat, we will be using data from the 1991 Survey of Income and Program Participation. According to the United States Census Website, the SIPP survey provides information on the features of economic well-being, including but not limited to education, wealth, and family structure. For the purpose of our analysis, we have included 7933 observations (rows) with data from persons ranging from 25-64 in age – the variable (feature) types are included in the next section. Then, we will be testing two different linear regression models to see which generates a more accurate and readable result of “total wealth” in US dollars, based on the data selected intuitively & statistically for training and testing purposes.

STATISTICAL ANALYSES

In order to predict total wealth in US dollars – our model's output – we have the following "feature variables" available:

All Predictors in 1991 SIPP Data (X)

ira: Individual Retirement Account with no employer matching (in US dollars)

e401: Binary variable; "1" if eligible for 401k, and "0" if not (only with participating employers)

nifa: Non 401k financial assets (in US dollars)

inc: Income (in US dollars)

hmort: Home mortgage payment (in US dollars)

hval: Value of home (in US dollars)

hequity: Home value minus home mortgage – hval minus hmort (in US dollars)

educ: Education (in years)

male: Binary variable; "1" if man, and "0" if not

twoearn: Binary variable; "1" if dual income household, "0" otherwise

nohs: Dummy variable; no high school diploma

hs: Dummy variable; only high school diploma

smcol: Dummy variable; some college

col: Dummy variable; some college degree

age: Age (in years)

fsize: Family size

marr: Binary variable; "1" if married, and "0" if not

Dependent Variable (Y)

tw: Total wealth (in US dollars)

Feature Variable Selection

In order to most accurately total wealth (Y), I believe that many of these variables should be incorporated as important predictors. However, many of these variables are collinear, so I must choose a larger variety to prevent this risk for collinearity. Additionally, I would like to drop any variables that are present from a construction of others (ie: hequity), as I don't want redundancy in my data. Lastly, it is clear that the variables used in the model's training and test sets must be intuitively relevant to our Y variable. The following remain after selection:

Model Predictors (X) – 14 in total

ira: Individual Retirement Account with no employer matching(in US dollars)

e401: Binary variable; “1” if eligible for 401k, and “0” if not (only with participating employers)

nifa: Non 401k financial assets (in US dollars)

inc: Income (in US dollars)

hmort: Home mortgage payment (in US dollars)

hval: Value of home (in US dollars)

educ: Education (in years)

male: Binary variable; “1” if man, and “0” if not

twoearn: Binary variable; “1” if dual income household, “0” otherwise

hs: Dummy variable; only high school diploma

col: Dummy variable; some college degree

age: Age (in years)

fsize: Family size

marr: Binary variable; “1” if married, and “0” if not

Notes on Selection:

- “hequity” has been taken out to prevent redundancy
- “nohs” and “smcol” have been taken out, as I do not believe they serve as relevant predictors
- ****Based on the EDA done in the “Feature Variable Inspection & Preliminary Findings”, additional variables may be dropped. The selections above are intuitive based on my knowledge of wealth accumulation and collinearity rules.**

MODEL SELECTION OUTLINE

We will be using and comparing the three following methods to see which holds a smaller mean square prediction error (MSPE). The one with the lowest MSPE represents our most accurate model of the two. The features of both models are contrasted below:

Ridge Regression (RR)

- Prevents multicollinearity
- Uses cross validation to achieve lambda (λ)
- Normally has lower MSPE/MSE than OLS (depends on the data)
- Not easy to interpret what exact features cause the outcome

Ridge Regression (RR) with Stepwise Feature Selection (SFS)

- Normally has lowest MSPE of the three
- Most complicated (more computation involved, and thus more room for error)
- Performs both statistical feature selection (SFS) and prediction (RR)

Ordinary Least Squares (OLS)

- Assumes no multicollinearity
- Easier to implement
- More interpretable with feature weighted outputs
- Assumes random sampling

FEATURE VARIABLE INSPECTION

After selecting my feature variables, I will now begin a preliminary analysis to examine how individual predictors (e.g., age, education) relate to total wealth in the United States. This analysis will help identify potential issues such as multicollinearity, outliers, and nonlinearity. Understanding these relationships will not only refine my selection of explanatory variables (X) but also guide necessary transformations to improve model performance and ensure a well-balanced training and test set.

BEGINNING CODE

First, I started by loading in the data:

```
setwd("/Users/mehrisadri/Downloads")
#Now, we will load the full dataset
data_tr <- read.table("data_tr.txt", header = TRUE, sep = "\t", dec = ".")[-1]
#Verifying that we loaded the correct dataset
head(data_tr)
```

In my preliminary selection, I decided to remove three different features:

```
#Now, I will remove some columns (available predictors) I don't want involved
data_tr <- data_tr[, !names(data_tr) %in% c("hequity", "nohs", "smcol")]
names(data_tr_fin)
#Now, I will get a general summary of the dataset before taking a closer inspect
# of certain variables to transform plain predictors so that our model
# runs smoother
summary(data_tr_fin)
```

Next, I decided to take a closer look at the tw variable and how it relates to some of our features. This can be done through a simple scatterplot using ggplot.

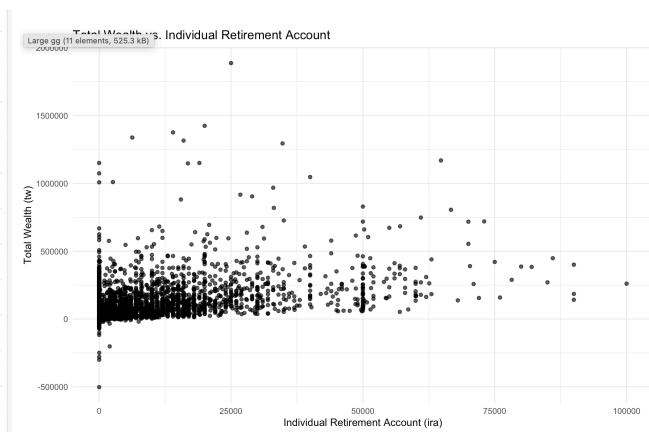
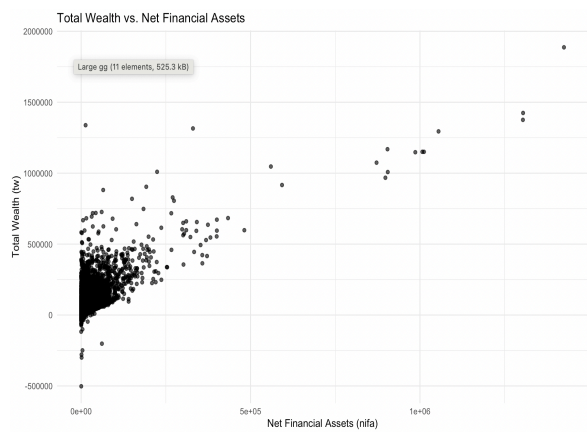
```
#Observing our Y value
hist(data_tr_fin$tw, breaks=100, main="histogram of Total Wealth", xlab="Total Wealth")
#Scatterplots
library(ggplot2)
plot(data_tr$tw, data_tr$age,
      xlab = "Total Wealth (USD)",
      ylab = "Age (in Years)",
      main = "Age vs. Total Wealth")
#We see above that there is a greater number of individuals ages 40 and up that have amassed greater wealth
#I have an intuitive understanding above how some variables change in relation to our Y
# value, but I am less sure about others. Thus, I will start with visualizing these
# nebulous datapoints
scatter_plot_hmort <- ggplot(data_tr, aes(x = hmort, y = tw)) +
  geom_point(alpha = 0.7) +
  theme_minimal() +
  labs(title = "Total Wealth vs. Home Mortgage",
       x = "Home Mortgage",
       y = "Total Wealth (USD)")
print(scatter_plot_hmort)
```

Scatterplot with Linear Fit



The scatter plot above highlights an unexpected: individuals with no home mortgage ($hmort = 0$) tend to have significantly higher total wealth compared to those with substantial mortgage payments. This is evident from the clusters of high-wealth individuals at both ends of the mortgage scale (circled in red). One possible explanation is that wealthier individuals may have fully paid off their homes, eliminating mortgage debt, while others with high mortgages may still be accumulating wealth but have large outstanding house debt.

I am less familiar with how ira (ira) and non-401(k) (nifa) assets factor into total wealth (output Y), since I don't have experience with either. Thus, I will visualize this relationship, too, with a scatterplot:



Clearly, the large number of observations makes it challenging to visualize a clear relationship between tw and ira/nifa. Thus, I will be performing statistical analysis for more precise selection.

FEATURE VARIABLE SELECTION

To better understand these features (non-graphically) and which ones should/shouldn't be incorporated into my final model, I will fit a linear regression model to assess how changes in ira and nifa impact tw.

Summary:

hmort Regression:

```
Call:
lm(formula = tw ~ hmort, data = data_tr)

Residuals:
    Min       1Q   Median       3Q      Max
-609930  -43247  -34423   14313  1789964

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 4.102e+04  1.464e+03   28.02  <2e-16 ***
hmort        7.484e-01  2.922e-02   25.61  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 104000 on 7931 degrees of freedom
Multiple R-squared:  0.07637,    Adjusted R-squared:  0.07625
F-statistic: 655.8 on 1 and 7931 DF,  p-value: < 2.2e-16
```

The wide range of residuals suggests prediction error when using only hmort in a linear regression model. The coefficient indicates that for every \$1 increase in total wealth, home mortgage value rises by approximately \$0.75. However, the relatively low R-squared value highlights that important predictors are still missing, making this model less reliable for wealth estimation.

nifa Regression:

```
Call:
lm(formula = tw ~ nifa, data = data_tr)

Residuals:
    Min       1Q   Median       3Q      Max
-597174  -43244  -27816   17173  1275449

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 4.349e+04  8.897e+02   48.88  <2e-16 ***
nifa        1.480e+00  1.665e-02   88.85  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 76630 on 7931 degrees of freedom
Multiple R-squared:  0.4989,    Adjusted R-squared:  0.4988
F-statistic: 7895 on 1 and 7931 DF,  p-value: < 2.2e-16
```

The positive and significant coefficient suggests that for every \$1 increase in net financial assets (nifa), total wealth rises by \$1.48. The substantially higher R-squared value compared to hmort indicates that nifa is a stronger predictor of total wealth. This suggests that financial liquidity – and not market assets – plays a more significant role in wealth accumulation than home mortgage obligations.

ira Regression:

```
Call:
lm(formula = tw ~ ira, data = data_tr)

Residuals:
    Min       1Q   Median       3Q      Max
-546492 -43440 -28190  15910 1702907

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.419e+04  1.121e+03   39.41  <2e-16 ***
ira          5.601e+00  1.098e-01   51.03  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 93920 on 7931 degrees of freedom
Multiple R-squared:  0.2472,    Adjusted R-squared:  0.2471
F-statistic: 2604 on 1 and 7931 DF,  p-value: < 2.2e-16
```

The model estimates that each \$1 increase in individual retirement accounts (ira) corresponds to a \$5.60 increase in total wealth, the highest impact among the three predictors. With an R-squared value of approximately 25%, ira explains more variance in total wealth than hmort but remains less predictive than nifa. This shows the strong influence of retirement savings on overall wealth but also indicates that additional factors must be considered for a more comprehensive model.

Overall, all three variables are *statistically significant*, so we will now examine their combined effect on total wealth through a multivariable, assuming no multicollinearity between them.

Regression with all three features (nifa, hmort, ira)

Call:

```
lm(formula = tw ~ nifa + hmort + ira, data = data_tr)
```

Residuals:

Min	1Q	Median	3Q	Max
-561924	-27898	-20326	11512	1278497

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.257e+04	9.719e+02	23.22	<2e-16 ***
nifa	1.252e+00	1.545e-02	81.04	<2e-16 ***
hmort	4.135e-01	1.927e-02	21.45	<2e-16 ***
ira	3.320e+00	8.331e-02	39.85	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 67530 on 7929 degrees of freedom

Multiple R-squared: 0.6109, Adjusted R-squared: 0.6107

F-statistic: 4149 on 3 and 7929 DF, p-value: < 2.2e-16

When analyzed together, ira remains the strongest predictor, with the steepest slope. The R-squared value of 0.61 indicates that these three variables explain a significant portion of total wealth. They will be retained in the final predictor list unless multicollinearity is detected.

FORWARD AND BACKWARDS REGRESSION (STEPWISE)

To further determine the most significant predictors, I will perform forward and backward stepwise regression. The two approaches will be: (1) starting with a model that includes all predictors and using backward elimination, and (2) starting with a model that includes no predictors (just the intercept) and using forward selection. This analysis was taught in Discussion 6.

Forward Summary:

Call:

```
lm(formula = tw ~ nifa + hval + hmort + ira + inc + e401 + age + male + marr, data = data_tr)
```

Residuals:

Min	1Q	Median	3Q	Max
-512079	-12915	-3245	4624	1279157

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.963e+04	2.254e+03	-8.708	< 2e-16 ***
nifa	1.062e+00	9.907e-03	107.168	< 2e-16 ***
hval	1.084e+00	1.059e-02	102.331	< 2e-16 ***
hmort	-1.047e+00	1.843e-02	-56.775	< 2e-16 ***
ira	1.625e+00	5.501e-02	29.545	< 2e-16 ***
inc	2.676e-01	2.588e-02	10.338	< 2e-16 ***
e401	7.500e+03	1.016e+03	7.383	1.7e-13 ***
age	3.243e+02	4.947e+01	6.555	5.9e-11 ***
male	3.702e+03	1.258e+03	2.943	0.00326 **
marr	-2.242e+03	1.117e+03	-2.007	0.04475 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 41560 on 7923 degrees of freedom

Multiple R-squared: 0.8527, Adjusted R-squared: 0.8526

F-statistic: 5097 on 9 and 7923 DF, p-value: < 2.2e-16

Backward Summary:

```
Call:
lm(formula = tw ~ ira + e401 + nifa + inc + hmort + hval + male +
    age + marr, data = data_tr)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-512079  -12915   -3245   4624 1279157
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.963e+04  2.254e+03  -8.708 < 2e-16 ***
ira          1.625e+00  5.501e-02  29.545 < 2e-16 ***
e401         7.500e+03  1.016e+03   7.383 1.7e-13 ***
nifa         1.062e+00  9.907e-03 107.168 < 2e-16 ***
inc          2.676e-01  2.588e-02  10.338 < 2e-16 ***
hmort        -1.047e+00  1.843e-02 -56.775 < 2e-16 ***
hval         1.084e+00  1.059e-02 102.331 < 2e-16 ***
male         3.702e+03  1.258e+03   2.943 0.00326 **
age          3.243e+02  4.947e+01   6.555 5.9e-11 ***
marr         -2.242e+03  1.117e+03  -2.007 0.04475 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 41560 on 7923 degrees of freedom
Multiple R-squared:  0.8527,    Adjusted R-squared:  0.8526
F-statistic: 5097 on 9 and 7923 DF,  p-value: < 2.2e-16
```

Since the variables *male* and *twoearn* are the least statistically significant, and both regression models yield similar significant results, I am inclined to remove them from the final predictor list.

COLLINEARITY AND MULTICOLLINEARITY

I am now testing *collinearity* and *multicollinearity* using regression with dummy variables – another technique taught during the discussion section. Since dummy variables are necessary for OLS regression, it is essential to convert categorical variables before proceeding.

```
data(Wage, package="ISLR")
# We transformed the matrix into a set of 0/1 dummy variables. Now every component is numeric!
new_X <- model.matrix(~, -1, data = Wage[, c(1:9)])
summary(Wage)
#Running regression model to identify multicollinear values
OLS <- lm(tw ~ ., data = data_tr)
summary(OLS)
```

OLS Output:

```
Call:
lm(formula = tw ~ ., data = data_tr)

Residuals:
    Min       1Q   Median       3Q      Max
-512449  -12927   -3258   4598 1279496

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.761e+04  4.336e+03  -4.061 4.92e-05 ***
ira          1.626e+00  5.526e-02  29.428  < 2e-16 ***
e401         7.544e+03  1.018e+03   7.413 1.37e-13 ***
nifa         1.062e+00  9.914e-03 107.081  < 2e-16 ***
inc          2.729e-01  2.737e-02   9.971  < 2e-16 ***
hmort        -1.047e+00  1.848e-02 -56.654  < 2e-16 ***
hval         1.085e+00  1.063e-02 102.060  < 2e-16 ***
educ         -9.184e+01  2.630e+02  -0.349  0.72697
male         3.658e+03  1.279e+03   2.860  0.00425 **
hs           -8.795e+02  1.079e+03  -0.815  0.41516
col          -7.571e+02  1.713e+03  -0.442  0.65854
age           3.170e+02  5.067e+01   6.257 4.13e-10 ***
fsize        -7.740e+01  3.826e+02  -0.202  0.83968
marr         -2.254e+03  1.302e+03  -1.731  0.08344 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 41570 on 7919 degrees of freedom
Multiple R-squared:  0.8527,    Adjusted R-squared:  0.8525
F-statistic: 3528 on 13 and 7919 DF,  p-value: < 2.2e-16
```

The regression output shows no evidence of multicollinearity, as there are no NA values and no concerning variance inflation issues among the predictors. Since none of the selected variables exhibit multicollinearity, I will proceed with the full set of predictors for OLS, RR, and SFS without any other modifications.

PART 1 CONCLUSION

Based on the analysis above, I will retain the variables I initially selected, except for *male* and *twoearn*, as the forward/backward regression results suggest they should be excluded.

Final Features Selected:

```
> data_tr_fin <- data_tr[, !names(data_tr) %in% c("twoearn", "man")]
> names(data_tr_fin)
 [1] "tw"      "ira"     "e401"    "nifa"    "inc"     "hmort"   "hval"    "educ"    "male"    "twoearn" "hs"
[12] "col"     "age"     "fsize"   "marr"
```

PREDICTION PERFORMANCE AND MODEL BUILDING

Ridge Regression only

Before building our Ridge Regression model first, it is important that we calculate the lambda parameter to build the most accurate analysis possible with the selected features.

I first will try a ridge regression using the features (variables) I selected through part 1's EDA as most significant and intuitively important to our output.

Using the R Commands taught, I will start by visualizing and then selecting the best lambda (λ) with my 15 feature variables. Our parameter will be between [-20, 20].

```
y <- data_tr$tw
X <- as.matrix(data_tr[,-1])

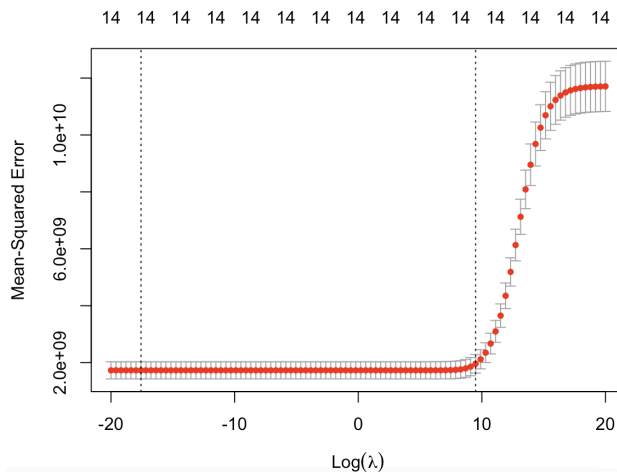
#Specifying the range (parameters) of Lambda -- [-20,20] for high variance
lambdas.rr <- exp(seq(-20, 20, length = 100))

set.seed(10)
```

Since we are performing a ridge regression, we want to make sure that alpha equals 0. If it were a lasso regression, we would set alpha = 1.

```
ridge_cv <- cv.glmnet(x = X, y = y, lambda = lambdas.rr, alpha = 0)
```

Then, we can get the optimal lambda and fit the model based on CV results. I will also plot it to fortify my results.



The optimal lambda value is found using the R command `ridge_cv$lambda.min`, and its effect on MSE is visualized with `plot(ridge_cv)`, confirming that MSPE remains lowest around 6 before increasing, making it a suitable choice to prevent over/underfitting. Note that we want a low MSPE.

Now that we've selected an ideal lambda value, I will properly train the data (80/20). The trained data with the values from CV will then be used to make predictions.

After this process, in lines 199-200 of my R script, I calculate the MSPE value, which measures the average squared difference between actual and predicted values. A lower MSPE (with $\lambda = 6$) indicates better model performance. Now, I will compare this MSPE with other regression methods to assess its effectiveness.

```
> mspe_ridge <- mean((y.test - pr.ridge)^2)
> print(paste("Ridge MSPE:", round(mspe_ridge, 4)))
[1] "Ridge MSPE: 2396093039.4404"
```

Ridge Regression vs Stepwise Regression

Now, we will perform k-fold cross-validation to compare stepwise regression (both forward and backward selection) with ridge regression – individually – for predicting total wealth. Stepwise regression selects features one-by-one, while ridge regression is applied with cross-validation to determine the optimal lambda for regularization after performing the stepwise. Similar to above, the mean squared prediction error (MSPE) is calculated for each approach to assess model performance.

```
# Stepwise Regression
full <- lm(tw ~ ., data = data_tr_fin[train, ])
null <- lm(tw ~ 1, data = data_tr_fin[train, ])
a <- stepAIC(null, scope = list(lower = null, upper = full), direction = 'forward')
b <- stepAIC(full, scope = list(lower = null, upper = full), direction = 'backward')
```

Comparing MSPE values across both methods after calculations:

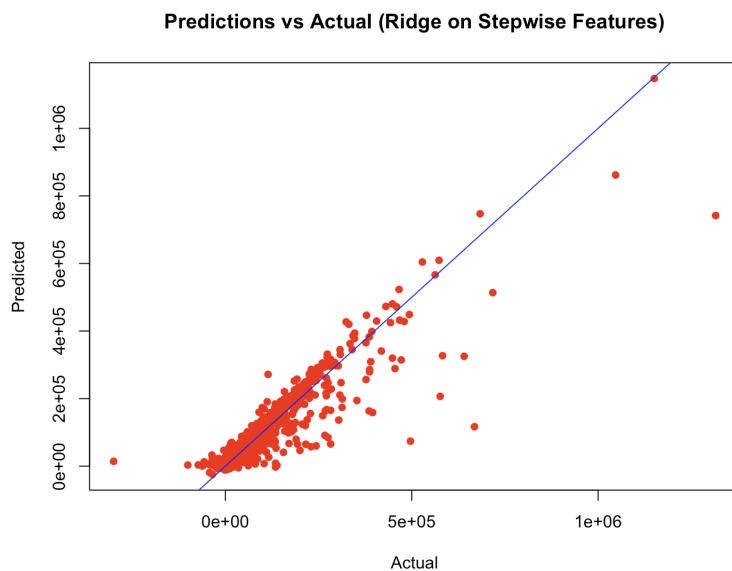
```
> #Comparing the MSPE values:
> print(mspe_step_backward) #Solely on backwards stepwise
[1] 1740214093
> print(mspe_step_forward) #Solely on forward stepwise
[1] 1740214093
> print(mspe_ridge) #Solely on ridge regression
[1] 1841206817
```

The MSPE values indicate that forward and backward stepwise regression resulted in the same prediction error, while ridge regression had a slightly higher MSPE of 1.84 billion. This suggests that stepwise regression has slightly better predictive performance than ridge regression.

Ridge Regression *with* Stepwise Regression

Now, let's do model prediction using both methods (one for feature selection, and the other for prediction):

```
> mspe_ridge_selected <- mean((y.test - pr.ridge_selected)^2)
> print(paste("Ridge MSPE on Stepwise Selected Features:", round(mspe_ridge_selected, 4)))
[1] "Ridge MSPE on Stepwise Selected Features: 1658189902.4314"
```



The mean squared prediction error (MSPE) values indicate that forward and backward stepwise regression produced identical results, both outperforming ridge regression (based on MSPE). To refine the model, a combination of stepwise regression for feature selection and ridge regression for regularization was applied, reducing the MSPE by a bit. This suggests that selecting key features before applying ridge regression enhances predictive performance. A visualization of actual vs. predicted values further evaluates the model's effectiveness. Additionally, the exclusion of certain variables like "male" and "marr" improved the model's accuracy according to the stepwise function, highlighting the importance of age, ira, home mortgage, and nifa predictors in getting the more accurate target variable.

Lastly, let's implement a simple ordinary least squares (OLS) regression for further comparison.

Ordinary Least Squares

```
Call:
lm(formula = tw ~ ., data = data_tr)

Residuals:
    Min       1Q   Median       3Q      Max
-516492 -12982   -3188    4818 1278869

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.761e+04  4.327e+03  -4.071 4.74e-05 ***
ira          1.616e+00  5.517e-02  29.294 < 2e-16 ***
e401         7.893e+03  1.018e+03   7.757 9.78e-15 ***
nifa         1.055e+00  9.954e-03 106.018 < 2e-16 ***
inc          3.155e-01  2.829e-02  11.149 < 2e-16 ***
hmort        -1.045e+00  1.844e-02 -56.669 < 2e-16 ***
hval         1.082e+00  1.062e-02 101.855 < 2e-16 ***
educ         -6.611e+01  2.625e+02  -0.252  0.80117
male         3.382e+03  1.277e+03   2.648  0.00812 **
twoearn      -7.542e+03  1.307e+03  -5.770 8.22e-09 ***
hs           -7.326e+02  1.077e+03  -0.680  0.49655
col          -1.153e+03  1.711e+03  -0.674  0.50029
age           2.897e+02  5.078e+01   5.704 1.21e-08 ***
fsize        -1.269e+02  3.819e+02  -0.332  0.73972
marr          1.781e+03  1.476e+03   1.207  0.22748
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 41480 on 7918 degrees of freedom
Multiple R-squared:  0.8534,    Adjusted R-squared:  0.8531
F-statistic: 3291 on 14 and 7918 DF,  p-value: < 2.2e-16
```

```
> #Computing MSE to compare with MSPE of Ridge, and MSPE of Stepwise + Ridge Regression
> MSE <- mean((data_tr$tw - ols_pred)^2)
> print(MSE)
[1] 1717619865
```

The regression output highlights key predictors of the target variable, with significant coefficients for IRA, employer-sponsored retirement plans (e401), net financial assets (nifa), income (inc), home mortgage (hmort), home value (hval), age, and two-income households (twoearn). Variables like education (educ), high school completion (hs), college education (col), and family size (fsize) appear to have minimal impact, which was also seen during part 1. The model achieves a high R-squared value of 0.8534, indicating strong explanatory power.

The ordinary least squares (OLS) model was applied to the training dataset, yielding a mean squared error (MSE) of approximately 1.72 billion. While this represents the lowest error among the models tested, there is still room for improvement. Comparing all MSPE values, the stepwise feature selection combined with ridge regression produced the most accurate predictions, demonstrating that regularization after feature selection enhances model performance.

GENERAL CONCLUSIONS

In summary, we started with setting our working directory and loading a dataset provided (1991 data). We then viewed the dataset, and removed certain variables from the dataset from intuitions on practicality and collinearity. Afterwards, I used R to generate a summary of the dataset to understand its basic statistics. Visualizations, like scatterplots, are created to explore the relationship between total wealth and variables such as age, home mortgage, and non-401k financial assets, including scatterplots and linear regression models to better understand the relationship between variable x and total wealth y . The stepwise regression, specifically, selects key predictors for total wealth (tw) by iteratively adding or removing variables based on their significance. This process identifies the most influential variables, improving model accuracy while avoiding overfitting. Overall, from the part 1 EDA, our models suggest that while home mortgage values has a minor effect on total wealth, non-401k financial assets *and* retirement accounts have a stronger impact in comparison. This conclusion is derived through observing R-squared values for these models indicate varying levels of predictive power, with net financial assets being a more significant predictor than home mortgage. Additionally, in part 1, we select what regression models we want to use for model testing and training.

There are three different methods discussed in part 2: RR, SFS, and OLS, each with their own advantages and disadvantages for total wealth prediction. In part 2, the ridge regression model is tuned using cross-validation (to find the ideal lambda value) and applied to predict total wealth, followed by testing its performance through mean squared prediction error (MSPE). Combining stepwise feature selection with ridge regression further enhances the model's accuracy, resulting in the lowest MSPE and improved predictive accuracy compared to other methods, including a simple OLS regression that was done at the end for comparison.

From this project, I was able to have hands-on experience to see all the factors involved with machine learning model prediction, using an economics lens. I learned that prediction accuracy is not only about the independent variables you select and the certain regression model pursued, but also how you transform the features through scaling, some of which is necessary to even pursue a regression model (ie: dummy variable creation).